

Ball annotation guide

How to label squash footage for the CrossCourt line-calling detector, and why each rule exists. Every number here was measured from the current dataset (SquashAI.coco, 1,367 frames / 1,103 annotations) on 2026-07-24.

The detector has one job: find a ball that is 4-25 px wide in a 4K frame. At that size the model is working near the limit of the available evidence, so labeling mistakes that would be harmless on a car or a person are fatal here.

1. The rule that matters most: never label consecutive frames

This is the single biggest thing to change. An audit of the current dataset:

Clip	Annotations	Median frame gap	Consecutive	Independent moments
ModelTrainTest2	309	1	95%	11
Bay Club Compilation	276	1	80%	14
SQUASHCLIP	167	1	89%	7
YouTube clip	161	1	67%	33
ModelTrainTest3	97	1	69%	17
IMG_3210	93	1	92%	5
Total	1,103			87

At 60 fps, frame n and frame $n+1$ share the same lighting, the same background, the same player positions, and a ball that has moved a few pixels. The network treats them as one example. **1,103 annotations are carrying roughly 87 samples' worth of information.**

The same labeling effort spread one frame per 30 would have produced ~1,100 distinct moments instead of 87 - more than ten times the information for identical cost.

Rule: minimum 30 frames between labels within a clip (0.5 s at 60 fps).

The only exception is the end-to-end eval set, where `eval_line_calls.py` needs contiguous frames around an impact to score an IN/OUT call. Keep those as short deliberate bursts, and keep them out of detector training data.

2. Label footage from the production rig

Camera and mount for anything destined for training or eval:

Setting	Value
Resolution / rate	4K, 60 fps, HEVC
Shutter	1/1000 s
ISO	Locked
White balance	Locked, set per court
Stabilization	OFF
Lens	Ultrawide
Mount	Top of back wall glass fins

Footage that does not match this is **pretraining data only**. It teaches the generic "small dark sphere against a court" feature, which transfers, but it cannot tell you how the product will behave.

The eval set must be 100% from the production rig, and frozen. You cannot improve what you cannot measure, and a score measured on off-rig footage predicts nothing. Aim for at least 3 different courts and 6 different sessions so you can report per-court variance rather than a single number that hides it.

Split by clip or session, never by frame. Because neighbouring frames are near-identical, a frame-level split puts copies of the same moment on both sides and reports a score that has nothing to do with generalisation.

3. Drawing the label

- **Use a box for most annotations.** At the measured median aspect ratio of 1.18 the ball is essentially round, and a polygon adds nothing over a box while taking several times longer to draw around a 9 px object.
- **Use a polygon only when the streak is visibly elongated** - roughly the 8% of annotations above aspect 2.0. There the major axis is real information: its two endpoints are where the ball was at the start and end of the exposure, which gives the tracker two timestamped positions from a single frame.
- **Draw tight to the ball**, and include the *whole* streak when there is one. Do not trim a streak back to a round blob - the extent is the measurement.
- **Label every ball you can actually see**, including partially occluded ones.
- **If you cannot see the ball, mark the frame as a negative.** Do not infer its position from the trajectory and place a box where it "must" be. A box on a patch of empty wall teaches the model to fire on empty wall - this is exactly what the old motion-blur augmentation was doing, and it cost roughly two thirds of the model's recall.
- **Never leave a frame unlabeled if a ball is visible.** An unlabeled visible ball is scored as a false positive during training, which actively teaches the model to suppress correct detections.

4. Negatives: budget 25-30% of your labeling time

Squash walls are covered in **ball marks** - dark smudges the exact size and colour of the ball, sitting on white plaster, hundreds per court. A detector trained to find "small dark blob on a light wall" will fire on every one of them. This is the single biggest false-positive source in this domain, and the only fix is showing the model a lot of them with no box attached.

Deliberately capture ball-free frames containing:

- Wall ball marks, especially dense clusters near the front wall
- Shoe soles and dark trainers
- Dark logos, sponsor boards, the tin
- Court line junctions and corners
- Shadows under players
- The ball held in a hand or pocket before a serve

5. Per-session diversity checklist

Spend the budget on variety, not volume. Tick each of these per session:

- ☐ **Court** - different wall colour, lighting, logos, floor tone
- ☐ **Ball depth** - front wall (smallest, hardest, and the one you must call), mid court, and near camera.
Over-sample front-wall balls relative to how often they occur.
- ☐ **Background** - white wall, glass back wall, floor, tin, across a player's body
- ☐ **Ball condition** - fresh and scuffed
- ☐ **Occlusion** - partially hidden by a player or racket
- ☐ **Speed** - hard drives and soft lobs
- ☐ **Negatives** - 25-30% of frames, per section 4

6. Never do

- Do not label runs of consecutive frames
- Do not place a box where you believe the ball is but cannot see it
- Do not skip a frame that has a visible ball

- Do not label footage below ~960 px wide - the ball is 3-4 px and there is no signal to learn
- Do not split train/val/test by frame
- Do not apply preprocessing or augmentation in Roboflow (see section 7)

7. Roboflow export settings

Generate every version with:

Preprocessing: Auto-Orient only. **No resize.** Never "Stretch to" - from 4K that is 7.5x horizontally against 4.2x vertically, which hands the network a round ball deformed into an ellipse.

Augmentation: none, and versions per training image set to 1. All augmentation belongs in the training loop where its magnitude can be scaled against the ball size.

Include unannotated images: unchecked.

Export format: COCO JSON - it carries real bbox fields alongside segmentation, so nothing has to infer boxes from polygons.

This is not a stylistic preference. The deployed v4 model was trained with `resize 512x512 "Stretch to"` and `motion-blur: 100 pixels` on a canvas where the median ball is 8.7 px. The blur erased the ball in about one of every three training copies while leaving the label attached.

8. Where to spend the next annotation budget

In priority order:

1. **~400 sparse frames for a frozen eval set** from the production rig, across at least 3 courts and 6 sessions. Never train on these. Highest value per label of anything on this list, because nothing else can be measured without it. 2. **~1,200 sparse frames for fine-tuning**, from the rig, from *different* sessions than the eval set. 3.

Active-learning rounds of ~300. After the first in-domain model exists, label where it fails rather than sampling randomly. Actively selected frames are worth several times their number in random ones.

Quick reference

Minimum gap between labels	30 frames (0.5 s at 60 fps)
Box or polygon	Box, unless the streak is visibly elongated
Ball visible but hard	Label it
Ball not visible	Negative - never a guessed box
Negative budget	25-30% of frames
Minimum source width	960 px
Split unit	Clip or session, never frame
Roboflow preprocessing	Auto-orient only
Roboflow augmentation	None
Export format	COCO JSON